

Product Information

Product Name

Single-Wavelength Pulse Oximeter

Revision: 1.14

General Description

Overview

The **Single-Wavelength Pulse Oximeter** is a compact, photoplethysmography (PPG)-based monitoring device designed to measure **heart rate (BPM)**. The system employs optical sensing combined with analog and digital signal processing to deliver real-time physiological data via an integrated microcontroller.

This device is **not a medical device** and must not be used for medical diagnosis, treatment, or clinical decision-making. Measurements are indicative only and not guaranteed to meet medical accuracy standards.

Applications

Primary Application

- Heart rate monitoring

Personal & Fitness Use

- Tracking resting heart rate and long-term trends
 - Monitoring heart rate changes during exercise
 - General fitness and wellness monitoring
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Electrical Specifications

Electrical Characteristics

Parameter	Specification
Supply Voltage	± 14 V DC
Analog Front-End	Low-noise op-amp amplification
Filtering	Active low-pass filtering
ADC Resolution	12-bit
LED Drive Voltage	6 V DC
LED Drive Current	40 - 50 mA

Operating Conditions

Parameter	Specification
Operating Temperature	-30 °C to $+70$ °C
Storage Temperature	0 °C to $+85$ °C
Ambient Light Rejection	None (no active filtering)
Intended Contact Area	Fingertip

Power Consumption

Parameter	Value
Typical Operating Current	20 - 65 mA
Power Suitability	Not suited for battery-powered operation

Performance Characteristics

Speed Metrics

Parameter	Value
Sampling Rate	15 Hz
Heart Rate Update Rate	1 Hz

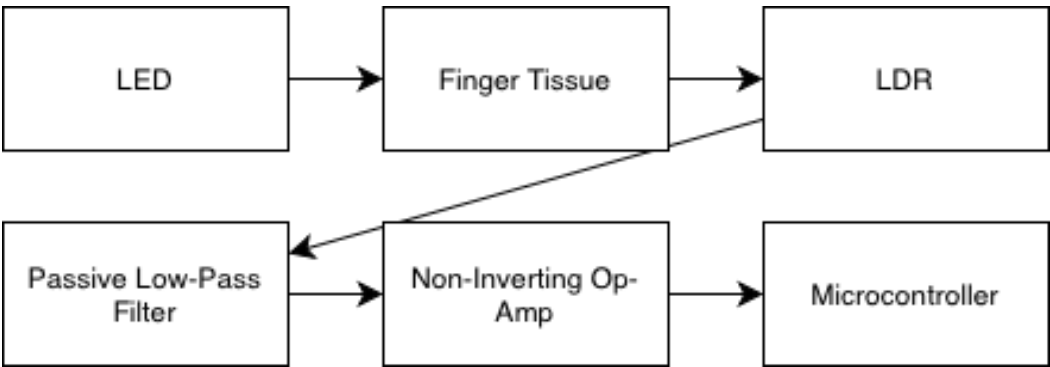
Accuracy Considerations

Heart rate accuracy is influenced by:

- Finger placement
- Contact pressure
- Ambient light conditions
- Motion artifacts

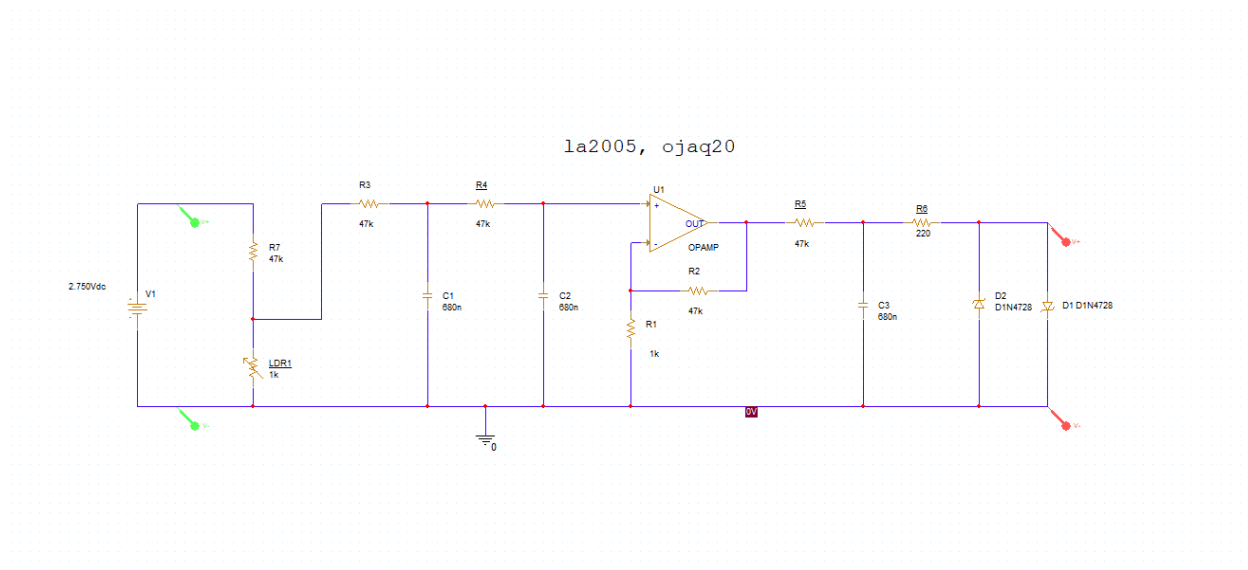
System Architecture

Block Diagram

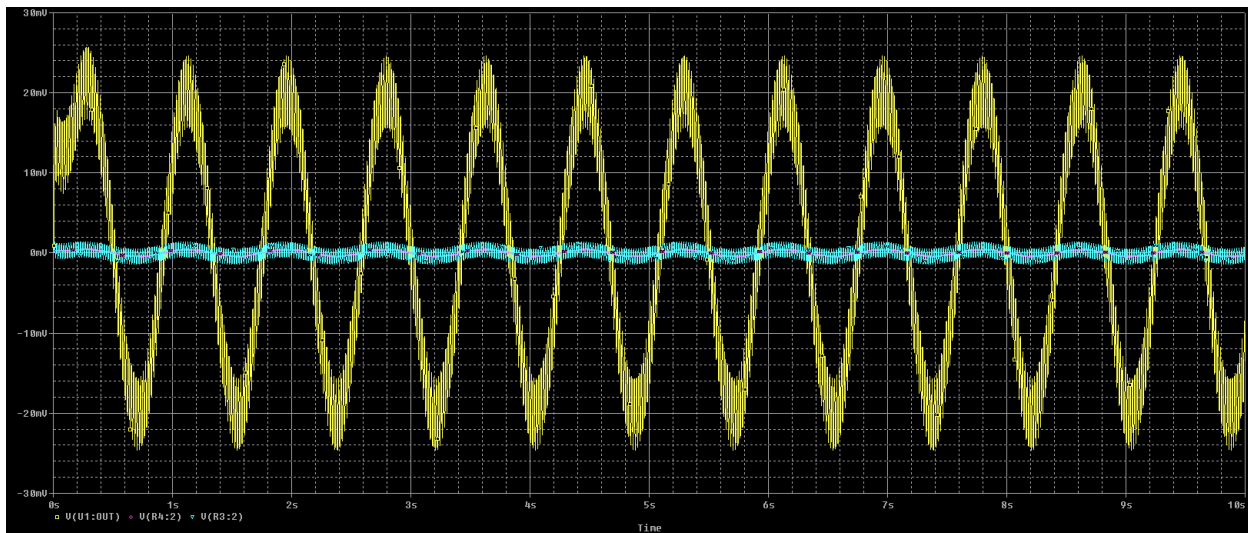


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Circuit Diagram



Simulated Output Waveform Post Filtering



Features

- Single-wavelength PPG sensing
- Low-noise analog front-end
- Passive band-pass filtering
- Real-time heart rate calculation
- Microcontroller-based signal processing
- Compact form factor optimized for fingertip measurements

Key Components

Component	Part Number	Manufacturer	Purpose	Datasheet
Light Dependent Resistor (LDR)	PGM5516	Token	PPG sensing	PGM5516
LED	MCL053MD	Multicomp Pro	Optical source for PPG	MCL053MD
Operational Amplifier	LM741	Texas Instruments	Signal amplification	LM741 OpAmp datasheet
Zener Diode (3.3 V)	1N4728A	Multicomp Pro	Voltage regulation	1N4728A
Resistors	—	Onecall	Biassing and gain setting	—
Capacitors	—	Onecall	Band-pass filtering	—
Microcontroller Board	Arduino Nano ESP32	Arduino	ADC and signal processing	Nano ESP32

Notes

- Device performance may vary depending on environmental and usage conditions.
- Future revisions may include active ambient light rejection and power optimization features.